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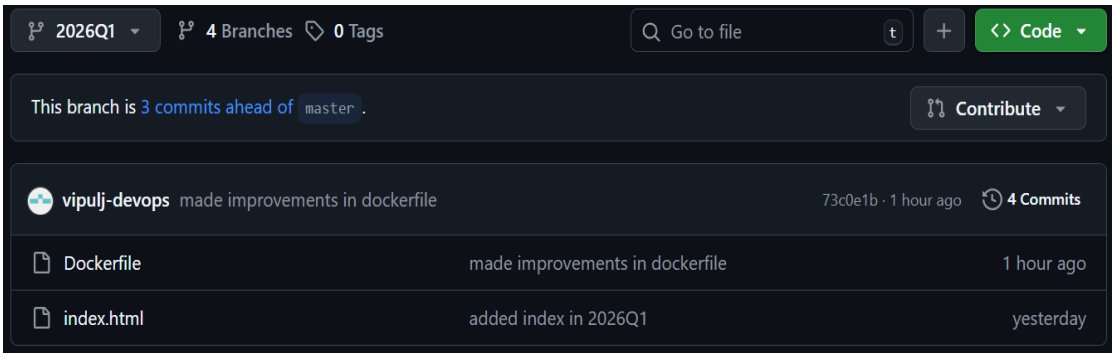
Assignment 4

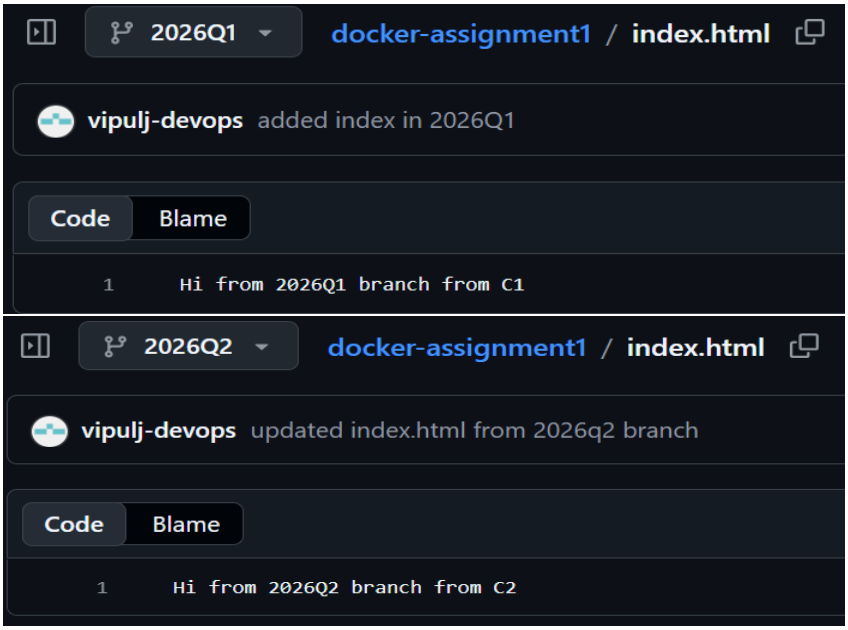
Task: Set up a Jenkins pipeline on the master node to build and deploy applications from a Git repository with branches 2026Q1, 2026Q2, and 2026Q3. The pipeline must run on two slave nodes in different availability zones. For each branch, build a Docker image and deploy an Apache HTTP server (httpd) container serving the respective index.html. Deploy containers c1, c2, and c3 on both nodes, exposing them on ports 80, 90, and 8080 (or another available port), respectively, with the entire process fully automated via the Jenkins pipeline. But now using volume in docker slave

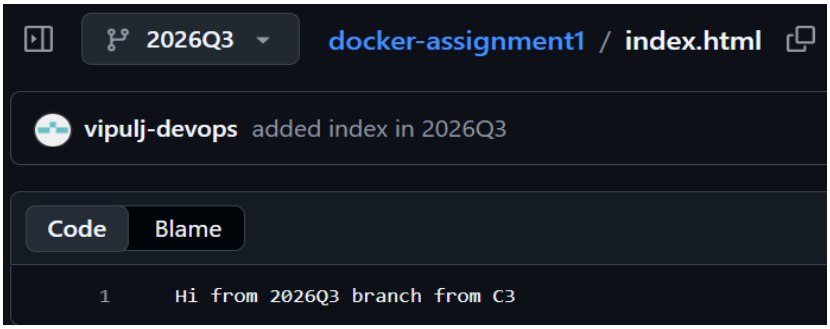
Steps:

1. Created a repository and added index.html and Dockerfile in it



2. Added index.html file in all 3 repositories





3. Added same Dockerfile in all 3 repositories

4. Launched Jenkins master and two slave instances

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
JM	i-010ff1fb4e340d912	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-13-126-232-254.ap...
slave1	i-070800d9972d6150a	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-13-232-123-154.ap...
slave2	i-024e4981c33b9e273	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-3-109-157-91.ap-s...

5. Installed java-17, docker, tomcat and Jenkins in Jenkins master

6. Installed java-17 and git in Jenkins slave

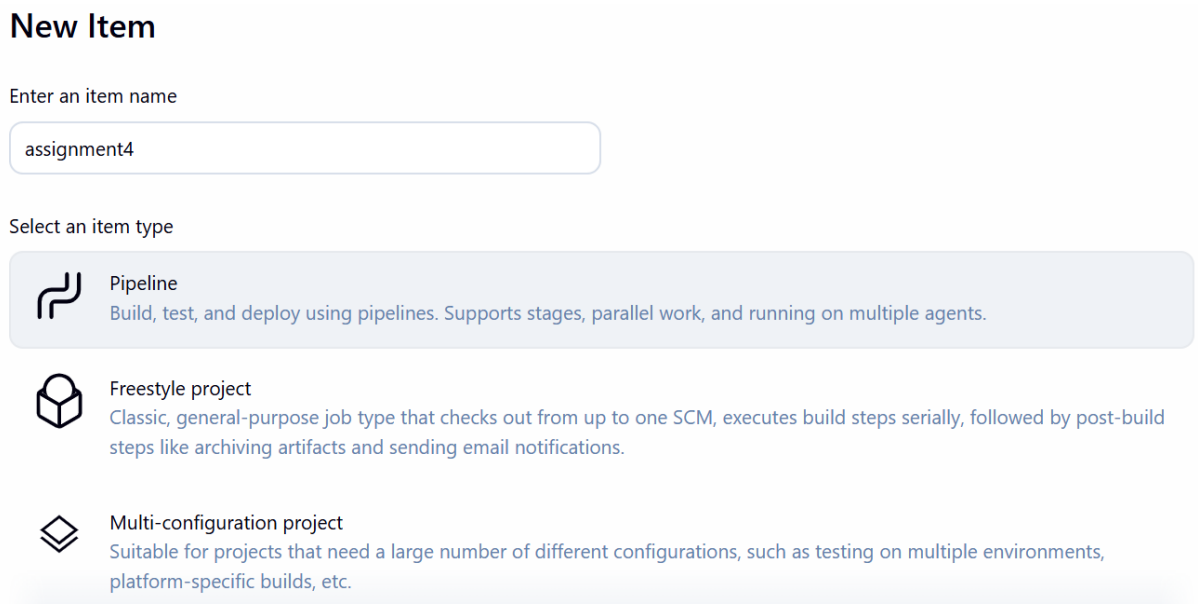
7. Created custom workspace directory as Jenkins slave in both slave instances and gave full permission to /mnt directory

```
[root@ip-172-31-43-131 jenkins-slave]# pwd
/mnt/jenkins-slave
```

8. Created Jenkins slave instances

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	5.31 GiB	0 B	5.31 GiB	0ms
	slave1	Linux (amd64)	In sync	5.30 GiB	0 B	458.38 MiB	36ms
	slave2	Linux (amd64)	In sync	5.58 GiB	0 B	458.37 MiB	27ms

9. Created declarative pipeline job on Jenkins master



10. Added Groovy script in the job and the job created successfully.

Script:

```
pipeline {
  agent none

  stages {

    stage('Deploy on Slave1') {
      agent {
        node {
          label 'slave1'
          customWorkspace '/mnt/jenkins-slave'
        }
      }

      stages {
```

```
stage('Clean') {
    steps {
        deleteDir()
        sh '''
        docker rm -f c1 c2 c3 || true
        docker system prune -a -f || true
        docker volume create deploy || true
        '''
    }
}

stage('Q1') {
    steps {
        git branch: '2026Q1',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'

        sh '''
        docker build -t image_q1 .

        sudo cp index.html /var/lib/docker/volumes/deploy/_data/
        sudo chmod -R 777 /var/lib/docker/volumes/deploy/_data

        docker run -d -p 80:80 --name c1 -v deploy:/mnt image_q1

        docker exec c1 cp -f /mnt/index.html /usr/local/apache2/htdocs/
        '''
    }
```

```
}

stage('Q2') {
    steps {
        deleteDir()

        git branch: '2026Q2',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'

        sh '''
        docker build -t image_q2 .

        sudo cp index.html /var/lib/docker/volumes/deploy/_data/
        sudo chmod -R 777 /var/lib/docker/volumes/deploy/_data

        docker run -d -p 90:80 --name c2 -v deploy:/mnt image_q2

        docker exec c2 cp -f /mnt/index.html /usr/local/apache2/htdocs/
        '''
    }
}

stage('Q3') {
    steps {
        deleteDir()

        git branch: '2026Q3',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'
```

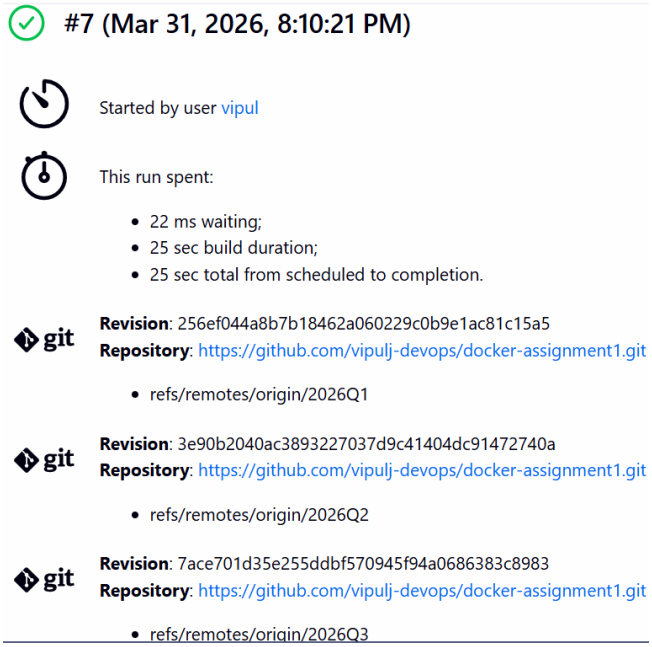
```
            sh '''
            docker build -t image_q3 .

            sudo cp index.html /var/lib/docker/volumes/deploy/_data/
            sudo chmod -R 777 /var/lib/docker/volumes/deploy/_data

            docker run -d -p 8081:80 --name c3 -v deploy:/mnt image_q3

            docker exec c3 cp -f /mnt/index.html /usr/local/apache2/htdocs/
            '''
        }
    }
  }
}

stage('Deploy on Slave2') {
  agent {
    node {
      label 'slave2'
      customWorkspace '/mnt/jenkins-slave'
    }
  }

  stages {

    stage('Clean') {
      steps {
```

```
        deleteDir()
        sh '''
        docker rm -f c1 c2 c3 || true
        docker system prune -f || true
        docker volume create deploy || true
        '''
    }
}

stage('Q1') {
    steps {
        git branch: '2026Q1',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'

        sh '''
        docker build -t image_q1 .

        sudo cp index.html /var/lib/docker/volumes/deploy/_data/
        sudo chmod -R 777 /var/lib/docker/volumes/deploy/_data

        docker run -d -p 80:80 --name c1 -v deploy:/mnt image_q1

        docker exec c1 cp -f /mnt/index.html /usr/local/apache2/htdocs/
        '''
    }
}

stage('Q2') {
```

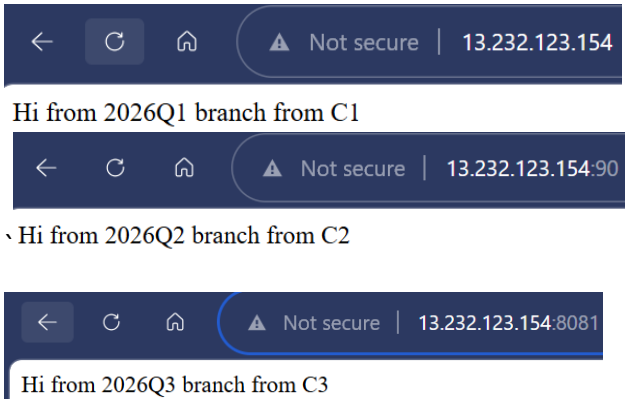
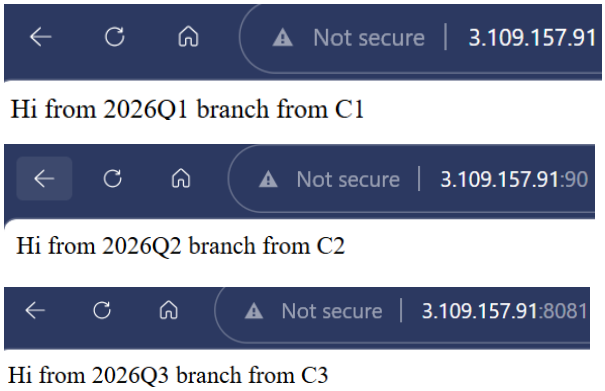
```
    steps {
        deleteDir()

        git branch: '2026Q2',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'

        sh '''
        docker build -t image_q2 .

        sudo cp index.html /var/lib/docker/volumes/deploy/_data/
        sudo chmod -R 777 /var/lib/docker/volumes/deploy/_data

        docker run -d -p 90:80 --name c2 -v deploy:/mnt image_q2

        docker exec c2 cp -f /mnt/index.html /usr/local/apache2/htdocs/
        '''
    }
}

stage('Q3') {
    steps {
        deleteDir()

        git branch: '2026Q3',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'

        sh '''
        docker build -t image_q3 .
```

```
            sudo cp index.html /var/lib/docker/volumes/deploy/_data/
            sudo chmod -R 777 /var/lib/docker/volumes/deploy/_data

            docker run -d -p 8081:80 --name c3 -v deploy:/mnt image_q3

            docker exec c3 cp -f /mnt/index.html /usr/local/apache2/htdocs/
            '''
        }
      }
    }
  }
 }
}
```

11. The pipeline ran successfully



12. Index.html deployed on Jenkins slave1



13. Index.html deployed on Jenkins slave2